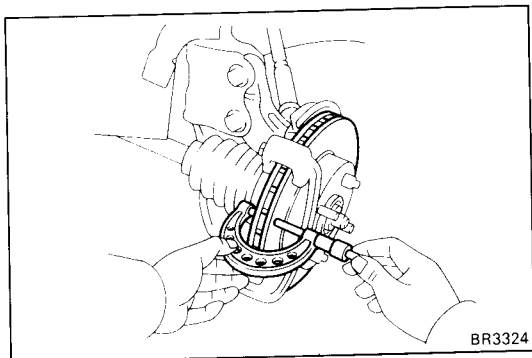


BRAKE SYSTEM

REFER TO 1990 CELICA REPAIR MANUAL (Pub. No. RM149U)

NOTE: The following pages contain only the points which differ from the above listed manual.

	Page
FRONT BRAKE	BR-2
PROPORTIONING VALVE (P VALVE)	BR-2
ANTI-LOCK BRAKE SYSTEM (ABS)	BR-3
Description	BR-3
Diagnosis System	BR-6
Troubleshooting	BR-10
Speed Sensor Diagnosis System	BR-16
Deceleration Sensor Operation	
Diagnosis System	BR-20
ABS Actuator	BR-22
Control Relay	BR-26
Front Speed Sensor	BR-27
Rear Speed Sensor	BR-29
Anti-Lock Brake System Circuit	BR-31



FRONT BRAKE

INSPECTION OF FRONT BRAKE COMPONENTS

MEASURE ROTOR DISC THICKNESS

Standard thickness: 25.0 mm (0.984 in.)

Minimum thickness: 23.0 mm (0.906 in.)

If the disc is scored or worn, or if thickness is less than minimum, repair or replace the disc.

PROPORTIONING VALVE (P VALVE)

INSPECTION OF FLUID PRESSURE

RAISE MASTER CYLINDER PRESSURE AND CHECK REAR WHEEL CYLINDER PRESSURE

Specifications

Master cylinder pressure	Rear wheel cylinder pressure
40 kg/cm ² (569 psi, 3,923 kPa)	40 kg/cm ² (569 psi, 3,923 kPa)
90 kg/cm ² (1,280 psi, 8,826 kPa)	58.5 kg/cm ² (832 psi, 5,737 kPa)

If the rear wheel cylinder pressure is incorrect, replace the P & B valve assembly.

ANTI-LOCK BRAKE SYSTEM (ABS)

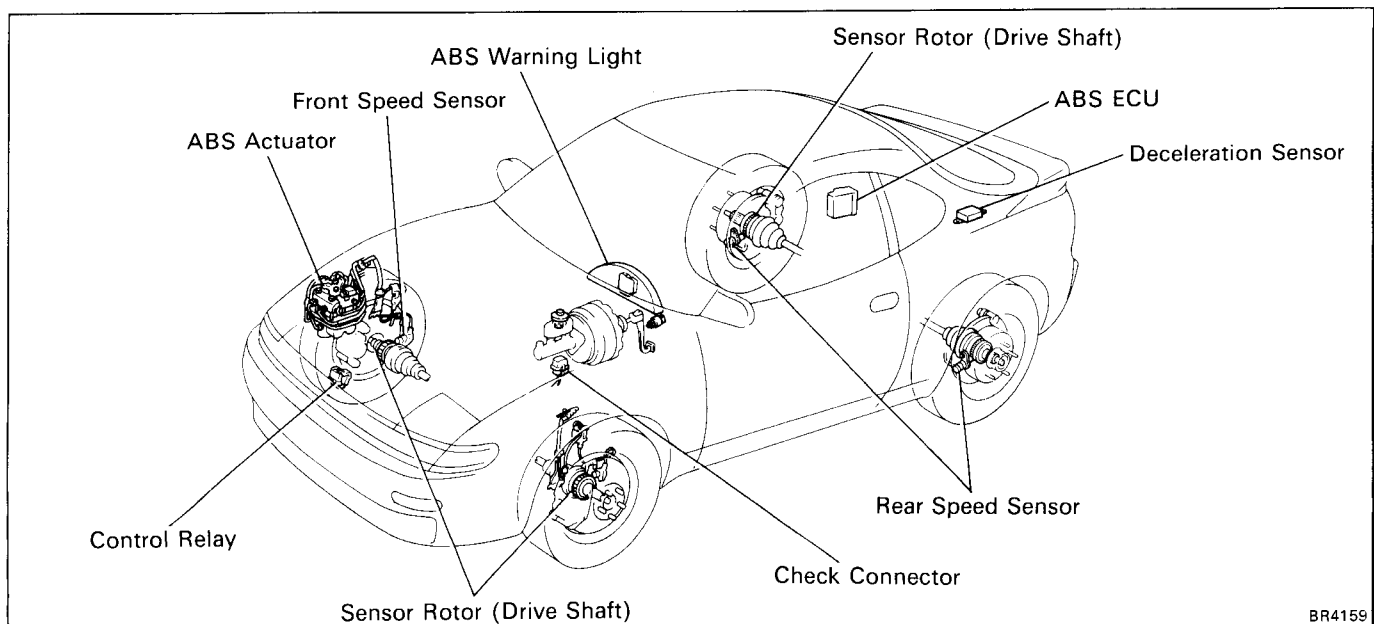
Description

- The ABS is a brake system which controls the wheel cylinder hydraulic pressure of all four wheels during sudden braking and braking on slippery road surfaces, preventing the wheels from locking. This ABS provides the following benefits:
 - (1) Enables steering round an obstacle with a greater degree of certainty even when panic braking.
 - (2) Enables stopping in a panic brake while keeping effect upon stability and steerability to a minimum, even on curves.
- The function of the ABS is to help maintain directional stability and vehicle steerability on most road conditions. However, the system cannot prevent the vehicle from skidding if the cornering speed limit is exceeded.
- The ABS has a longitudinal deceleration sensor to match braking characteristics to the full-time four-wheel drive.
- In case a malfunction occurs, a diagnosis function and fail-safe system have been adopted for the ABS to increase serviceability.

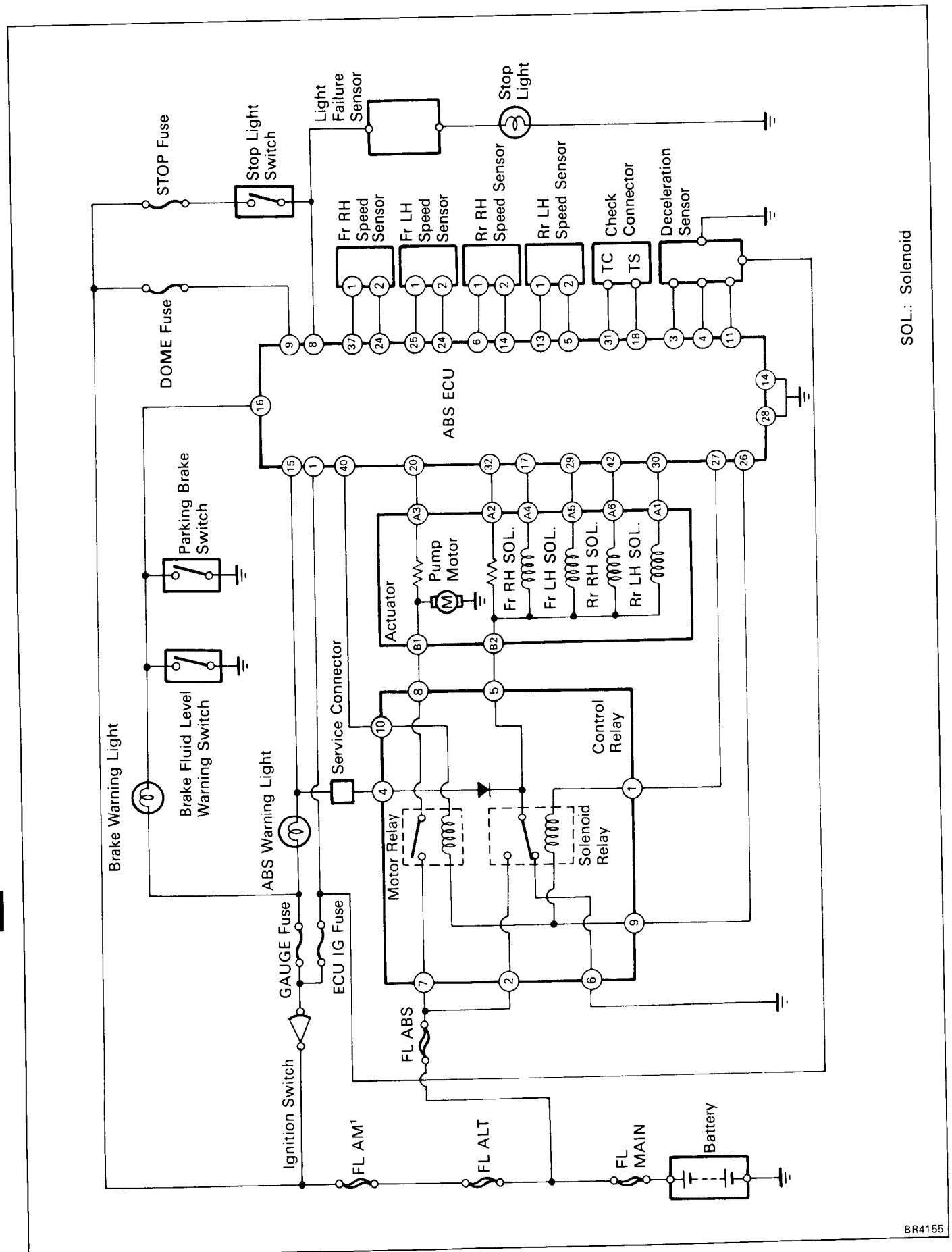
FUNCTION OF COMPONENTS

Component	Function
Front Speed Sensor	Detect the wheel speed of each of the left and right front wheels.
Rear Speed Sensor	Detect the wheel speed of each of the left the right rear wheels.
Deceleration Sensor	Detects the deceleration speed of the vehicle and sends a signal accordingly to the ABS ECU.
ABS Warning Light	Lights up to alert the drive when trouble has occurred in the Antilock Brake System.
Actuator	Controls the brake fluid pressure to each disc brake cylinder through signals from the computer.
ABS ECU	From the wheel speed signals from each sensor, it calculates acceleration, deceleration and slip values and sends signals to the actuator to control brake fluid pressure.

LOCATION OF SYSTEM PARTS

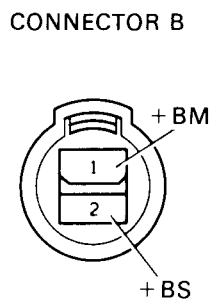
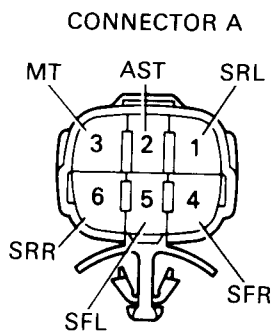


WIRING DIAGRAM

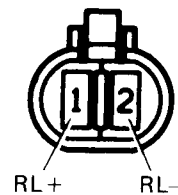


CONNECTORS

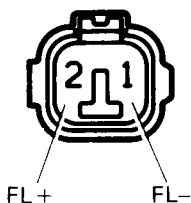
ABS Actuator



Rear Speed Sensor (LH)



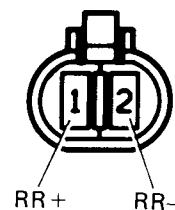
Front Speed Sensor (LH)



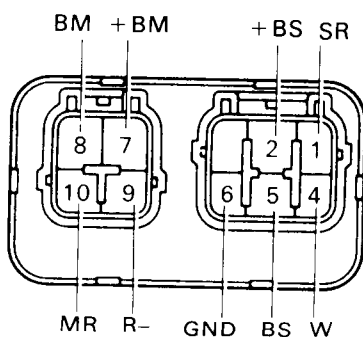
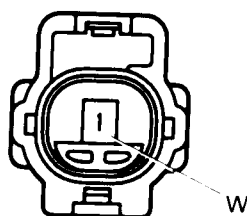
Front Speed Sensor (RH)



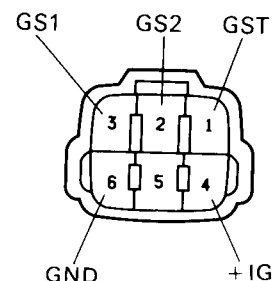
Rear Speed Sensor (RH)



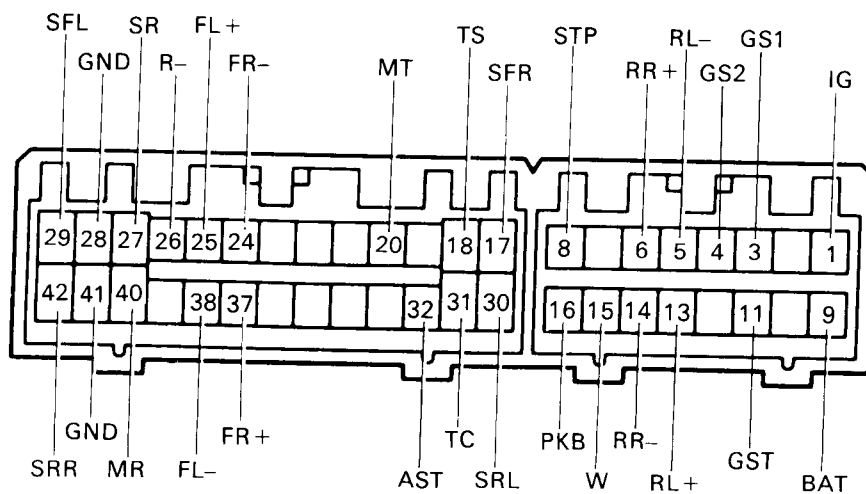
Control Relay

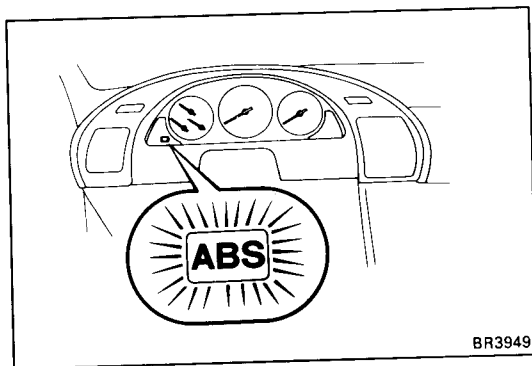
Service Connector
(Wire Harness Side)

Deceleration Sensor

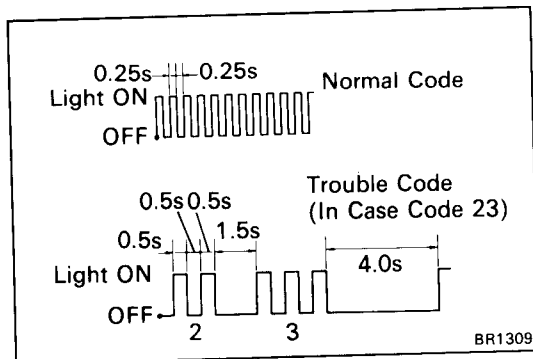


ABS ECU





BR3949



BR1309

Diagnosis System

DESCRIPTION

If a malfunction occurs, the system will identify the problem and the ECU will store the codes for the trouble items.

At the same time, the system informs the driver of a malfunction via the "ABS" warning light in the combination meter.

To identify the trouble by the number of blinks (diagnostic code) of the warning light turn on the ignition switch, disconnect the service connector, and use SST to connect Tc and E1 of the check connector.

In the event of two codes, that having the smallest number (code) will be identified first.

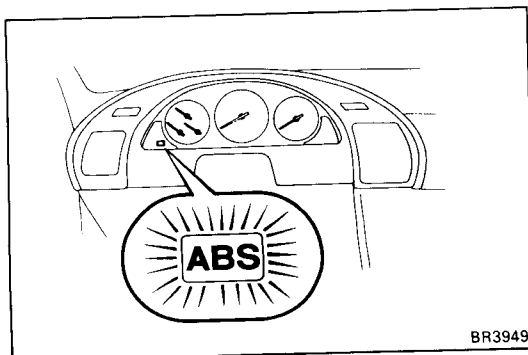
HINT: The warning light does not show the diagnostic codes while the vehicle is running.

INITIAL CHECK

CHECK ACTUATOR OPERATION NOISE

- Start the engine and drive at a speed over 6 km/h (4 mph).
- Check that the actuator operation noise is heard.

HINT: A initial check is carried out once each time after the engine has been started and initial speed exceeds 6 km/h (4 mph). The respective functions, in order, of the 3 position solenoid and pump motor in the actuator are checked. However, if the brake pedal is depressed, the initial check is not carried out, but is started after the pedal has been released.



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INSPECTION OF DIAGNOSIS SYSTEM

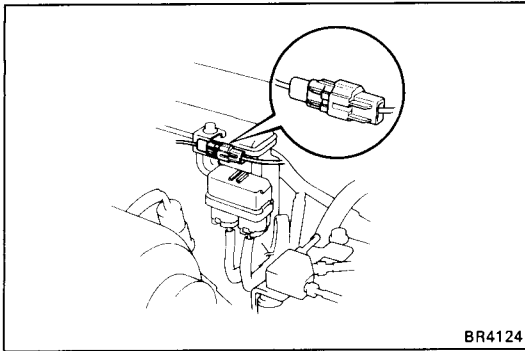
1. INSPECT BATTERY VOLTAGE

Inspect that the battery voltage is about 12 V.

2. CHECK THAT WARNING LIGHT TURNS ON

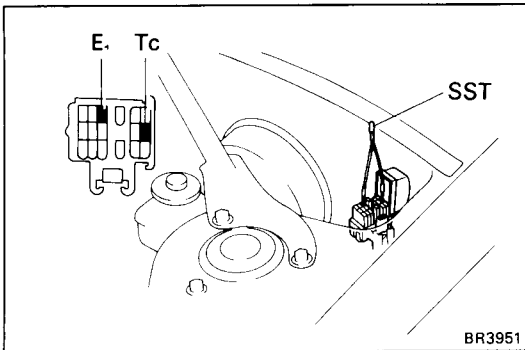
- Turn the ignition switch on.
- Check that the "ABS" warning light turns on for 3 seconds.

If not, inspect and repair or replace the fuse, bulb and wire harness.



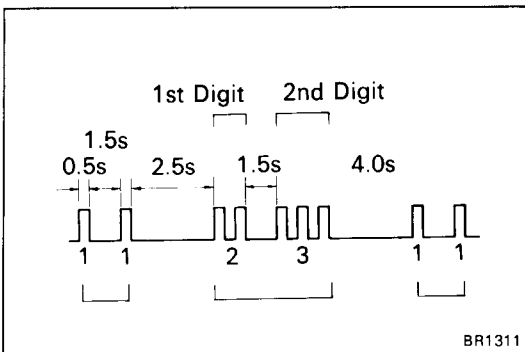
3. READ DIAGNOSTIC CODE

- (a) Turn the ignition switch on.
- (b) Disconnect the service connector.



- (c) Using SST, connect terminals Tc and E1 of the check connector.

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- (d) In event of a malfunction, 4 seconds later the warning light will begin to blink. Read the number of blinks.

(See DIAGNOSTIC CODE on page BR-8)

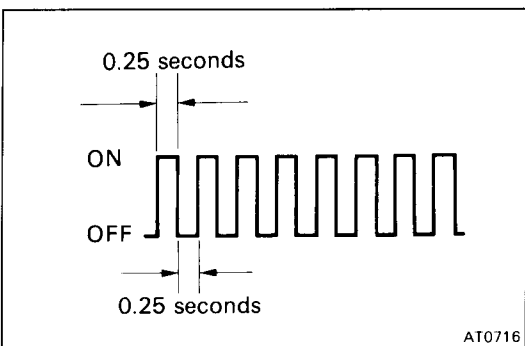
HINT: The first number of blinks will equal the first digit of a two digit diagnostic code. After a 1.5 second pause, the 2nd number of blinks will equal the 2nd number of a two digit code. If there are two or more codes, there will be a 2.5 second pause between each, and indication will begin after 4.0 second pause from the smaller value and continue in order to larger.

- (e) If the system is operating normally (no malfunction), the warning light will blink once every 0.5 seconds.

- (f) Repair the system.

- (g) After the malfunctioning components has been repaired, clear the diagnostic codes stored in the ECU. (See page BR-9)

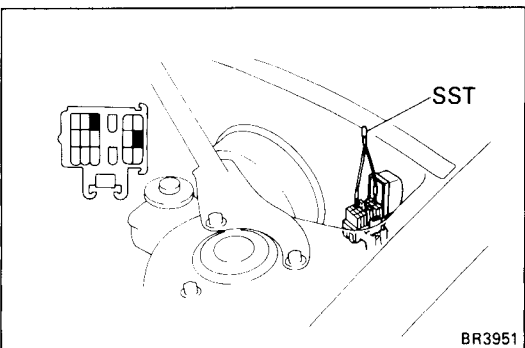
HINT: If you disconnect the battery cable while repairing, all diagnostic codes in the ECU will erased.

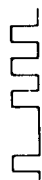
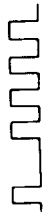
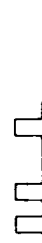
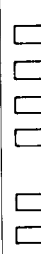
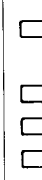
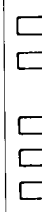
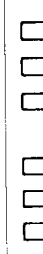
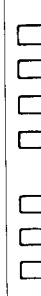
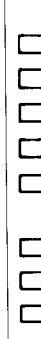
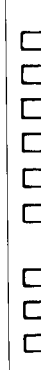
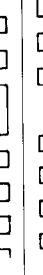


- (h) Remove the SST from terminals Tc and E1 of the check connector.

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- (i) Connect the service connector.
- (j) Turn the ignition switch on, and check that the "ABS" warning light goes off after the warning light goes on for 3 seconds.



Code No.	Light Pattern	Diagnosis	Trouble Part
11	ON OFF 	Open circuit in solenoid relay circuit	• Actuator inside wire harness • Control relay • Wire harness and connector of solenoid relay circuit
12		Short circuit in solenoid relay circuit	
13		Open circuit in pump motor relay circuit	• Actuator inside wire harness • Control relay • Wire harness and connector of pump motor relay circuit
14		Short circuit in pump motor relay circuit	
21		Open or short circuit in 3 position solenoid of front right wheel	
22		Open or short circuit in 3 position solenoid of front left wheel	
23		Open or short circuit in 3 position solenoid of rear right wheel	
24		Open or short circuit in 3 position solenoid of rear left wheel	
31		Front right wheel speed sensor signal malfunction	• Actuator solenoid • Wire harness and connector of actuator solenoid circuit
32		Front left wheel speed sensor signal malfunction	
33		Rear right wheel speed sensor signal malfunction	• Speed sensor • Sensor rotor • Wire harness and connector of speed sensor
34		Rear left wheel speed sensor signal malfunction	
35		Open circuit in front left or rear right wheel speed sensor	
36		Open circuit in front right or rear left wheel speed sensor	
41		Abnormal battery voltage (9.5 V less than or 16.2 V more than)	• Battery • Voltage regulator
43		Malfunction in deceleration sensor	• Deceleration sensor • Deceleration sensor installation • Wire harness and connector of deceleration sensor
44		Open or short circuit in deceleration sensor	
51		Pump motor of actuator locked or open circuit in pump motor circuit in actuator	• Pump motor, relay and battery • Wire harness, connector and ground bolt or actuator pump motor circuit
Always on		Malfunction in computer	• ECU

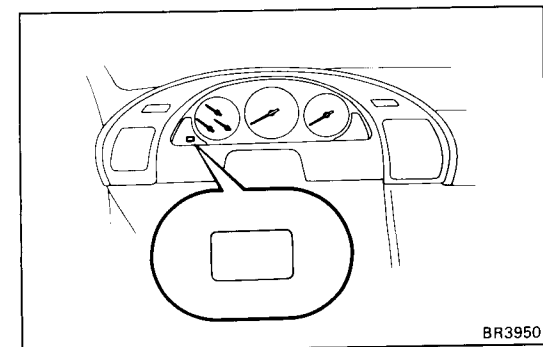
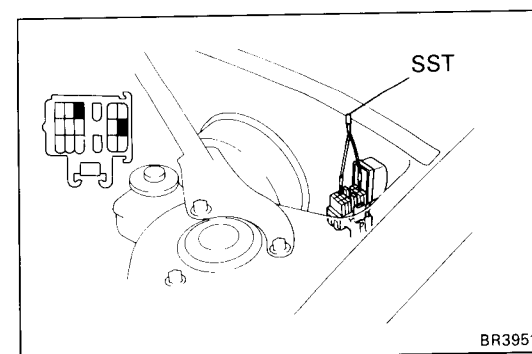
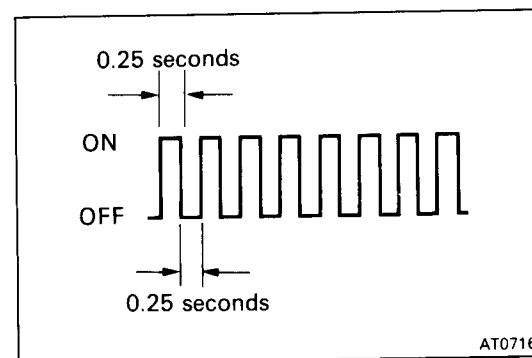
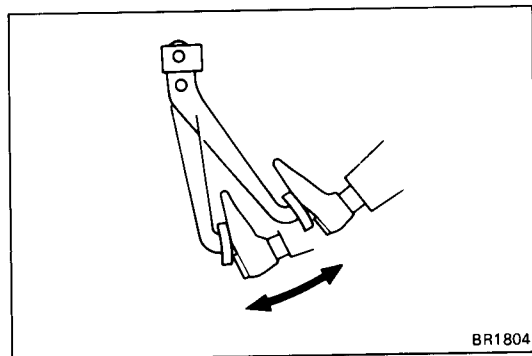
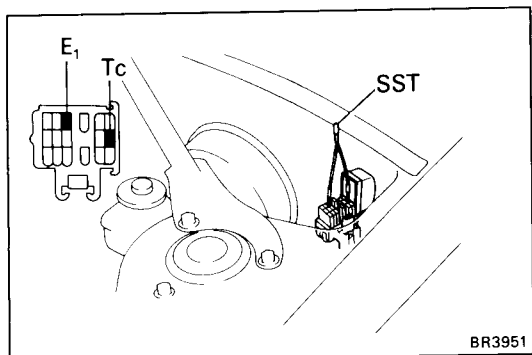
CLEARING OF DIAGNOSTIC CODES

CLEAR DIAGNOSTIC CODES

- (a) Turn the ignition switch on.
- (b) Using SST, connect terminals Tc and E1 of the check connector.

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HINT: Keep the vehicle stopped vehicle speed 0 km/h (0 mph).



- (c) Clear the diagnostic codes stored in ECU by depressing the brake pedal 8 or more times within 3 seconds.

- (d) Check that the warning light shows the normal code.

- (e) Remove the SST from terminals Tc and E1 of the check connector.

SST 09843-18020

- (f) Check that the warning light goes off.

Troubleshooting

Problem		No.
"ABS" warning light	Always comes on after ignition switch is turned on.	1
	Does not come on for 3 seconds after ignition switch on.	2
	Comes on and off.	3
	Comes on while running.	1
Brake working	Brakes pull.	4
	Braking inefficient.	4
	ABS operates at ordinary braking.	4
	ABS operates just before stopping at ordinary braking.	4
	Brake pedal pulsates abnormally while ABS is operating.	4
	Skidding noise occurs while ABS working. (ABS works inefficiently)	5

1

"ABS" warning light comes on.

Disconnect service connector and connect terminals Tc and E₁ of check connector.
(See page BR-7)

Does warning light always come on or show the normal code? (Ignition switch on)

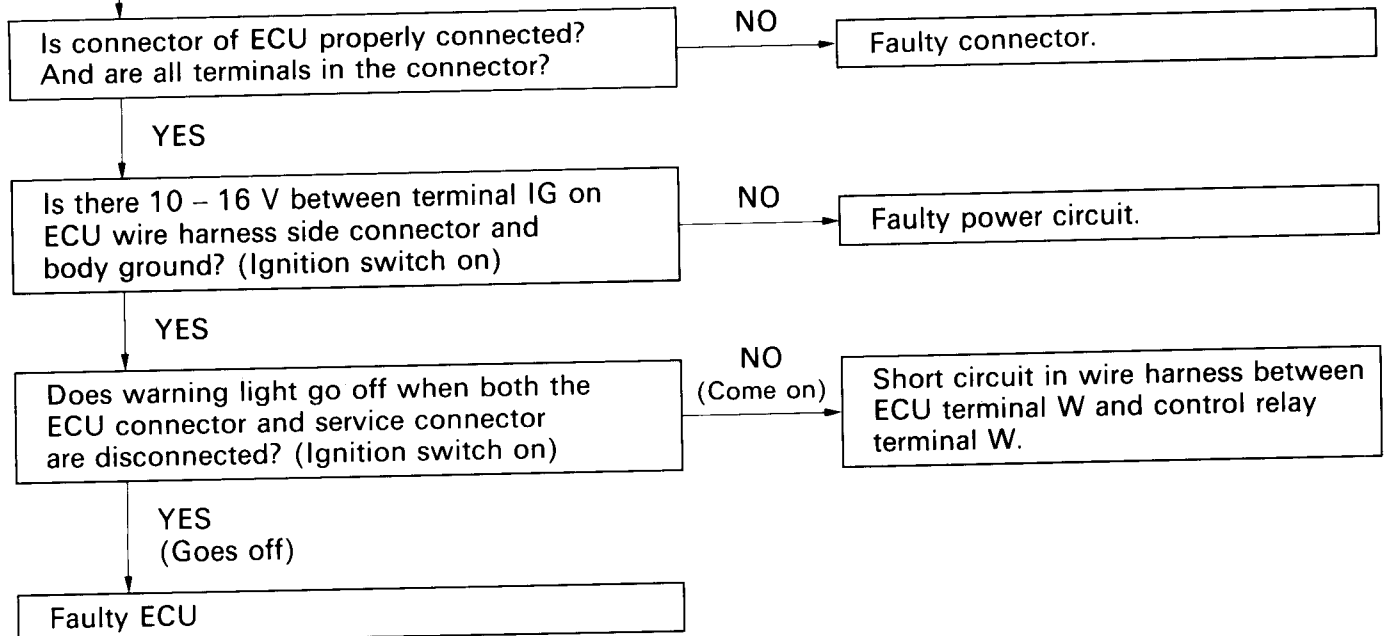
NO

See diagnostic code.
(See page BR-8)

YES

Continued on page BR-11

Continued from page BR-10



2

"ABS" warning light does not come on for 3 seconds after ignition switch on.

Disconnect service connector, and ground the terminal W of the female side connector. (Ignition switch on)

Does warning light come on?

NO

Bulb burned out or open circuit in wire harness between warning light and control relay terminal W.

YES

Disconnect connectors from ECU and control relay, and ground terminal W of ECU wire harness side connector. (Ignition switch on)

Does warning light come on?

NO

Open circuit in wire harness between computer terminal W and warning light.

YES

With ignition switch off, disconnect control relay connectors, and check continuity between terminal W and BS on actuator side.

Reverse tester leads and check again. Is there one way continuity between terminals?

NO

Short circuit in control relay inside diode.

YES

Faulty ECU.

HINT: If the diode is short-circuited, a malfunction at ECU terminal W will occur. When inspecting the terminal, connect the ECU connector, and disconnect control relay connectors and service connector. Then turn the ignition switch on, and check that the warning light goes on. If it does, the ECU terminal is OK.

3 "ABS" warning light comes on and off.

- Check for short circuit in wire harness between terminal Tc and E₁ or Ts and E₁ of check connector.

4

- Brakes pull.
- Braking inefficient.
- ABS operates at ordinary braking.
- ABS operates just before stopping at ordinary braking.
- Brake pedal pulsates abnormally while ABS working.

Disconnect service connector and connect terminals Tc and E₁ of check connector.
(See page BR-7)

Does warning light show the diagnostic normal code? (Ignition switch on)

NO

See diagnostic code.
(See page BR-8)

YES

Try speed sensor diagnosis system.
Is sensor signal level OK?
(See page BR-16)

NO

Inspect speed sensor, and replace if necessary.

YES

Try speed sensor diagnosis system.
Is sensor signal change OK?
(See page BR-16)

NO

Inspect sensor rotor, and replace if necessary.

YES

Are each speed sensors installed in place?
And are each installation bolts tightened securely?

NO

Speed sensor installation faulty.

YES

Is there foreign material or ferric chips on the sensor tip?

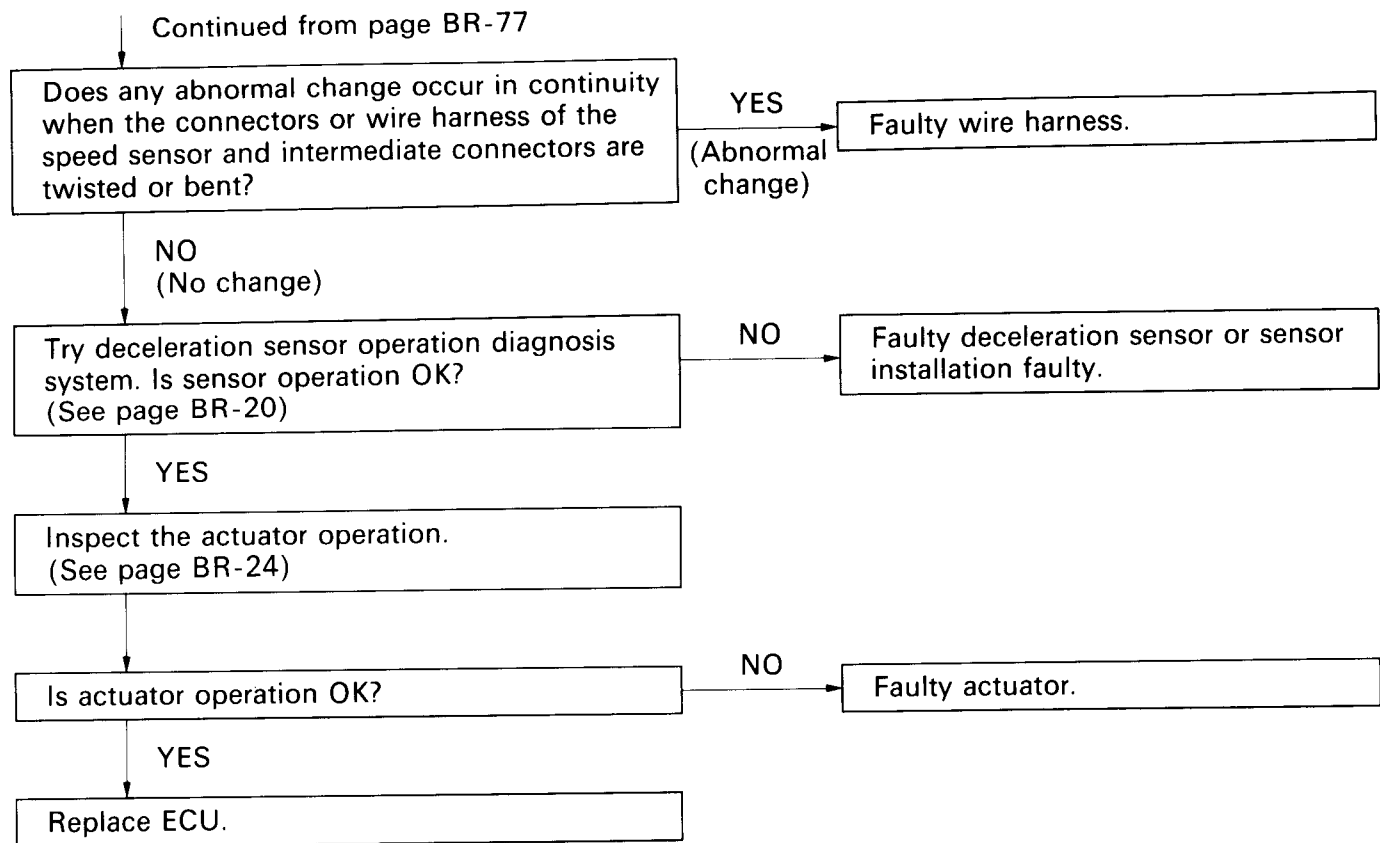
YES

Clean chips from the speed sensor.

NO

Disconnect connector from ECU, inspect continuity between each speed sensor terminals on wire harness side.
(See page BR-32)

Continued on page BR-14



- 5 Antilock brake system works inefficiently.

Disconnect service connector and connect terminals Tc and E₁ of check connector.
(See page BR-7)

Does warning light show the diagnostic normal code? (Ignition switch on)

NO

See diagnostic code.
(See page BR-8)

YES

Is there battery voltage between ECU terminal STP and body ground when depressing brake pedal?

NO

Open circuit in stop light switch and/or wire harness.

YES

Inspect actuator.
(See page BR-24)

Speed Sensor Diagnosis System

PRECAUTION

While checking the speed sensor diagnosis system, ABS does not work and brake system works as normal brake system.

INSPECTION OF DIAGNOSIS SYSTEM

1. INSPECT BATTERY VOLTAGE

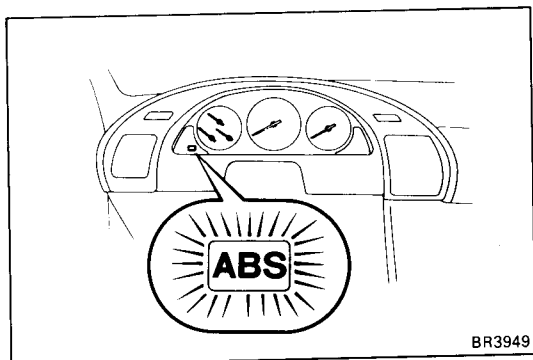
Inspect that the battery voltage is about 12 V.

2. CHECK THAT WARNING LIGHT TURNS ON

- (a) Turn the ignition switch on.
- (b) Check that the "ABS" warning light turns on for 3 seconds.

If not, inspect and repair or replace the fuse, bulb and wire harness.

- (c) Check that the "ABS" warning light turns off.
- (d) Turn the ignition switch off.



BR3949

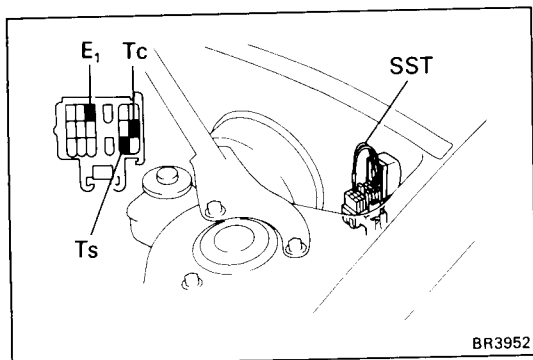
3. PERFORM FOLLOWING STEPS

- (a) Using SST, connect terminals E1 and Tc, Ts of the check connector.

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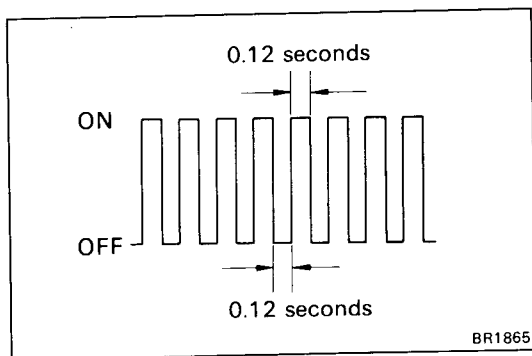
- (b) Pull the parking brake lever up, and start the engine.

HINT: Do not depress the brake pedal.

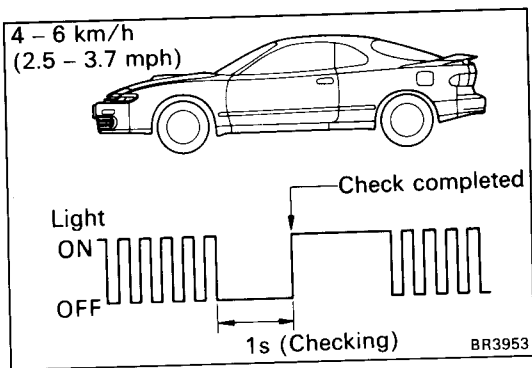


BR3952

- (c) Check that the warning light blinks about 4 times every 1 second as shown.



BR1865



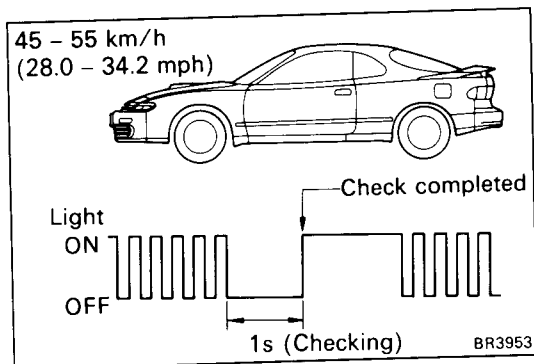
4. INSPECT SENSOR SIGNAL LEVEL

Drive the vehicle straight ahead at about 4 – 6 km/h (2.5 – 3.7 mph), and check that the warning light turns on after a 1 second pause.

If the warning light turns on without blinking when the vehicle speed is not within the specified speed range above, stop the vehicle and read the diagnostic code, and repair the malfunctioning parts.
(See step 6 on this page)

HINT: If the warning light turns on while the vehicle speed is within specified speed range above, the check is completed. And when the vehicle speed exceeds 6 km/h (3.7 mph), the warning light will blink again. In this condition, speed sensors are OK.

NOTICE: While the warning light is off, do not give any shocks to vehicle such as acceleration, deceleration, braking, shift change, steering or shocks from the road condition.



5. INSPECT SENSOR SIGNAL CHANGE

Drive the vehicle straight ahead at about 45 – 55 km/h (28.0 – 34.2 mph), and check that the warning light turns on after a 1 second pause.

If the warning light turns on without blinking when the vehicle speed is not within the specified speed range above, stop the vehicle and read the diagnostic code, and repair the malfunctioning parts.
(See step 6 on this page)

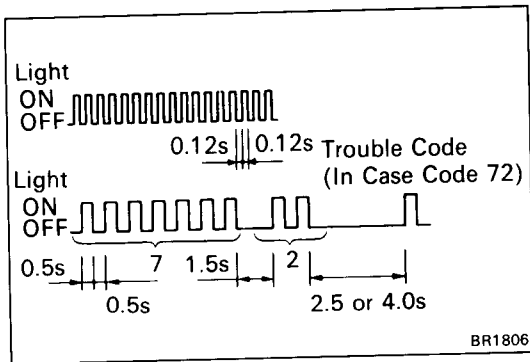
HINT: If the warning light turns on while the vehicle speed is within specified speed range above, the check is completed. And when the vehicle speed is not within specified speed range, the warning light will blink again. In this condition, sensor rotors are OK.

NOTICE: While the warning light is off, do not give any shocks to vehicle such as acceleration, deceleration, braking, shift change, steering or shocks from the road condition.

6. READ DIAGNOSTIC CODE

Stop the vehicle, and warning light will begin to blink.
Read the number of blinks.
(See DIAGNOSTIC CODE on page BR-19)

NOTICE: Do not depress the brake pedal more than 16 times after the diagnosis system begins to work, or the diagnostic codes stored in ECU will be cleared.



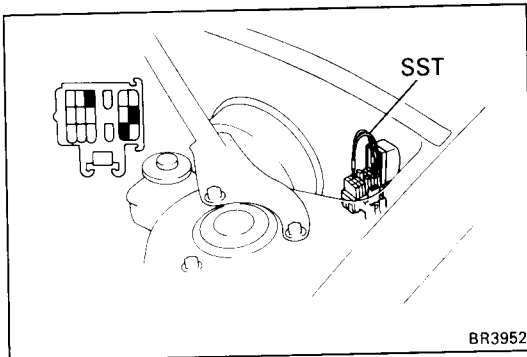
HINT: The first number of blinks will equal the first digit of a two digit diagnostic code. After a 1.5 second pause, the 2nd number of blinks will equal the 2nd number of a two digit code. If there are two or more codes, there will be a 2.5 second pause between each code, and then indication will begin again after a 4.0 second pause, continuing in order from the smaller value up to the larger one.

7. REPAIR MALFUNCTIONING PARTS

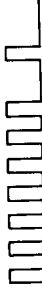
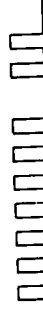
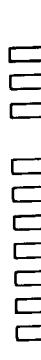
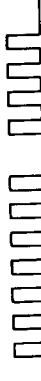
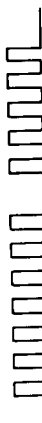
Repair or replace the malfunctioning parts.

8. PERFORM FOLLOWING STEPS

- (a) Turn the ignition switch off.
- (b) Remove the SST from terminals E1 and Tc, Ts of the check connector.



DIAGNOSTIC CODE

Code No.	Light Pattern	Diagnosis	Malfunctioning Part
	ON  OFF	All speed sensors and sensor rotors are normal	
71		Low voltage of front right speed sensor signal	• Front right speed sensor • Sensor installation
72		Low voltage of front left speed sensor signal	• Front left speed sensor • Sensor installation
73		Low voltage of rear right speed sensor signal	• Rear right speed sensor • Sensor installation
74		Low voltage of rear left speed sensor signal	• Rear left speed sensor • Sensor installation
75		Abnormal change of front right speed sensor signal	• Front right sensor rotor
76		Abnormal change of front left speed sensor signal	• Front left sensor rotor
77		Abnormal change of rear right speed sensor signal	• Rear right sensor rotor
78		Abnormal change of rear left speed sensor signal	• Rear left sensor rotor

Deceleration Sensor Operation Diagnosis System

PRECAUTION

While checking the deceleration sensor diagnosis system, ABS does not work and brake system works as normal brake system.

INSPECTION OF DIAGNOSIS SYSTEM

1. INSPECT BATTERY VOLTAGE

Inspect that the battery voltage is about 12 V.

2. CHECK THAT WARNING LIGHT TURNS ON

- Turn the ignition switch on.
- Check that the "ABS" warning light turns on for 3 seconds.

If not, inspect and repair or replace the fuse, bulb and wire harness.

- Check that the "ABS" warning light turns off.
- Turn the ignition switch off.

3. PERFORM FOLLOWING STEPS

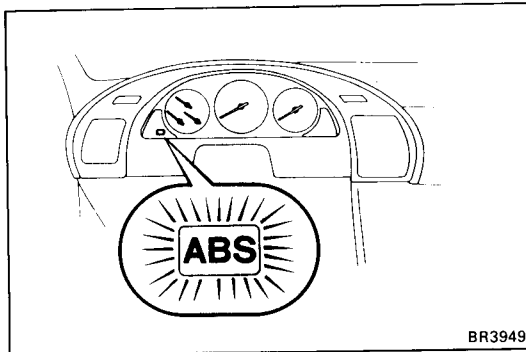
- Using SST, connect terminals Ts and E1 of the check connector.

SST 09843-18020

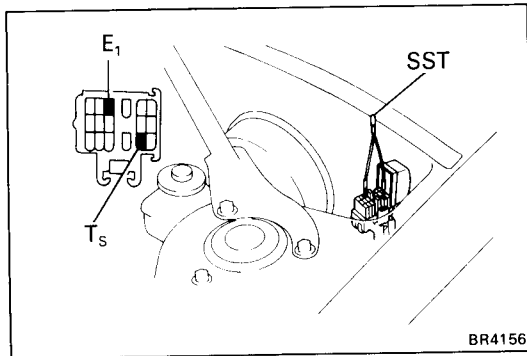
- Pull the parking brake lever up and depress the brake pedal, and start the engine.

- Check that the warning light blinks about 1 time every 1 second as shown.

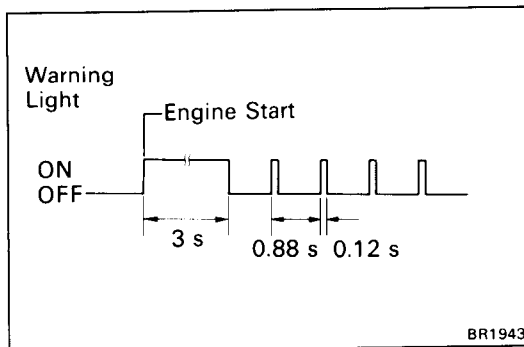
If the warning light does not blink, inspect the parking brake switch, stop light switch, Ts connector, deceleration sensor installation and computer.



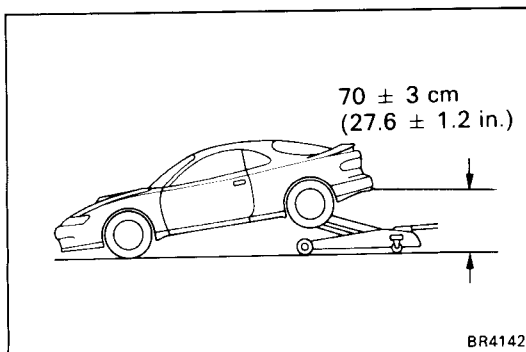
BR3949



BR4156



BR1943



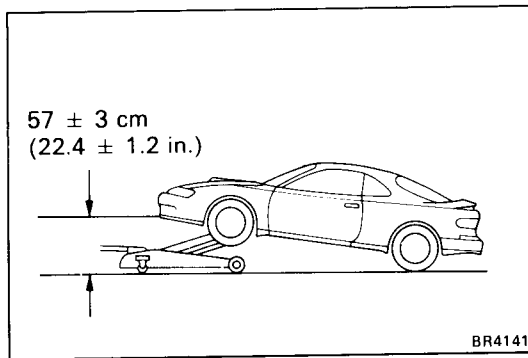
BR4142

4. INSPECT SENSOR DETECTION POINT

- Jack up the rear side of the vehicle slowly as shown.

HINT: When measuring the height, measure at the center of the rear bumper lower edge.

- Check that the warning light does not turn on.
- Jack down the vehicle and check that the warning light blinking.



(d) Jack up the front side of the vehicle slowly as shown.

HINT: When measuring the height, measure at the center of the lower body or spoiler edge of the vehicle.

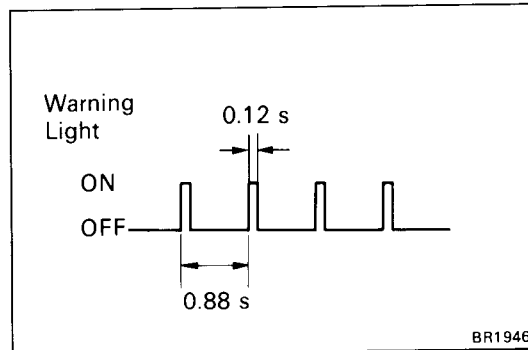
(e) Check that the warning light does not turn on.

(f) Jack down the vehicle and check that the warning light blinking.

If the warning light turns on, inspect the deceleration sensor installation. And if the sensor installation is OK, replace the deceleration sensor.

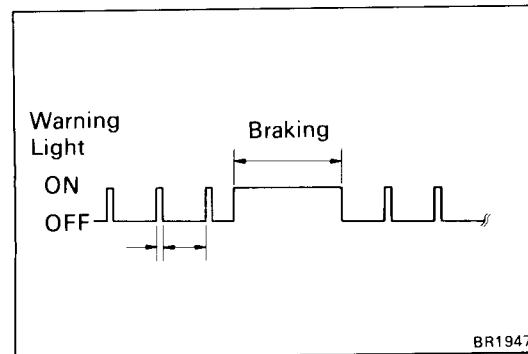
5. INSPECT SENSOR OPERATION

- Drive the vehicle straight ahead at about 10 km/h (6.2 mph) or more, lightly depress the brake pedal.
- Check that there is no change in the warning light pattern.



- Drive the vehicle straight ahead at about 20 km/h (12.4 mph) or more, depress the brake pedal a little strong.

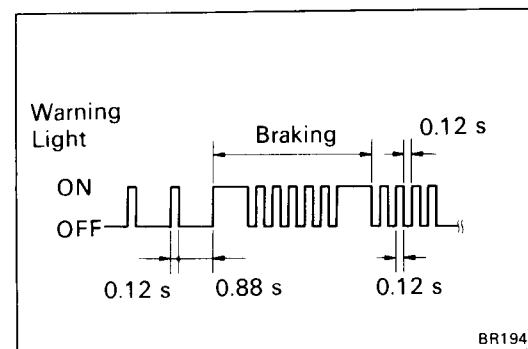
(d) Check that the warning light turns on while braking.



- Drive the vehicle straight ahead at about 20 km/h (12.4 mph) or more, depress the brake pedal strongly.

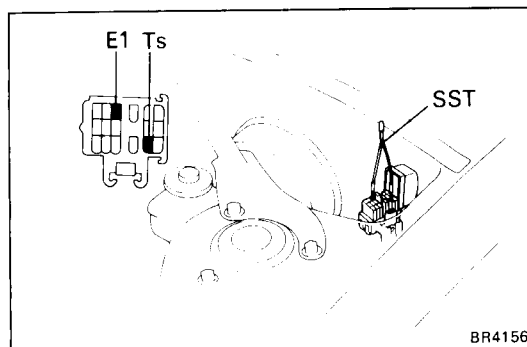
(f) Check that the warning light light pattern changes after braking as shown.

If the operation is not as specified, inspect the deceleration sensor installation. And if the sensor installation is OK, replace the deceleration sensor.



6. PERFORM FOLLOWING STEPS

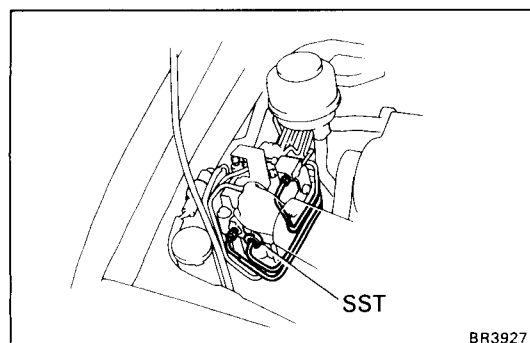
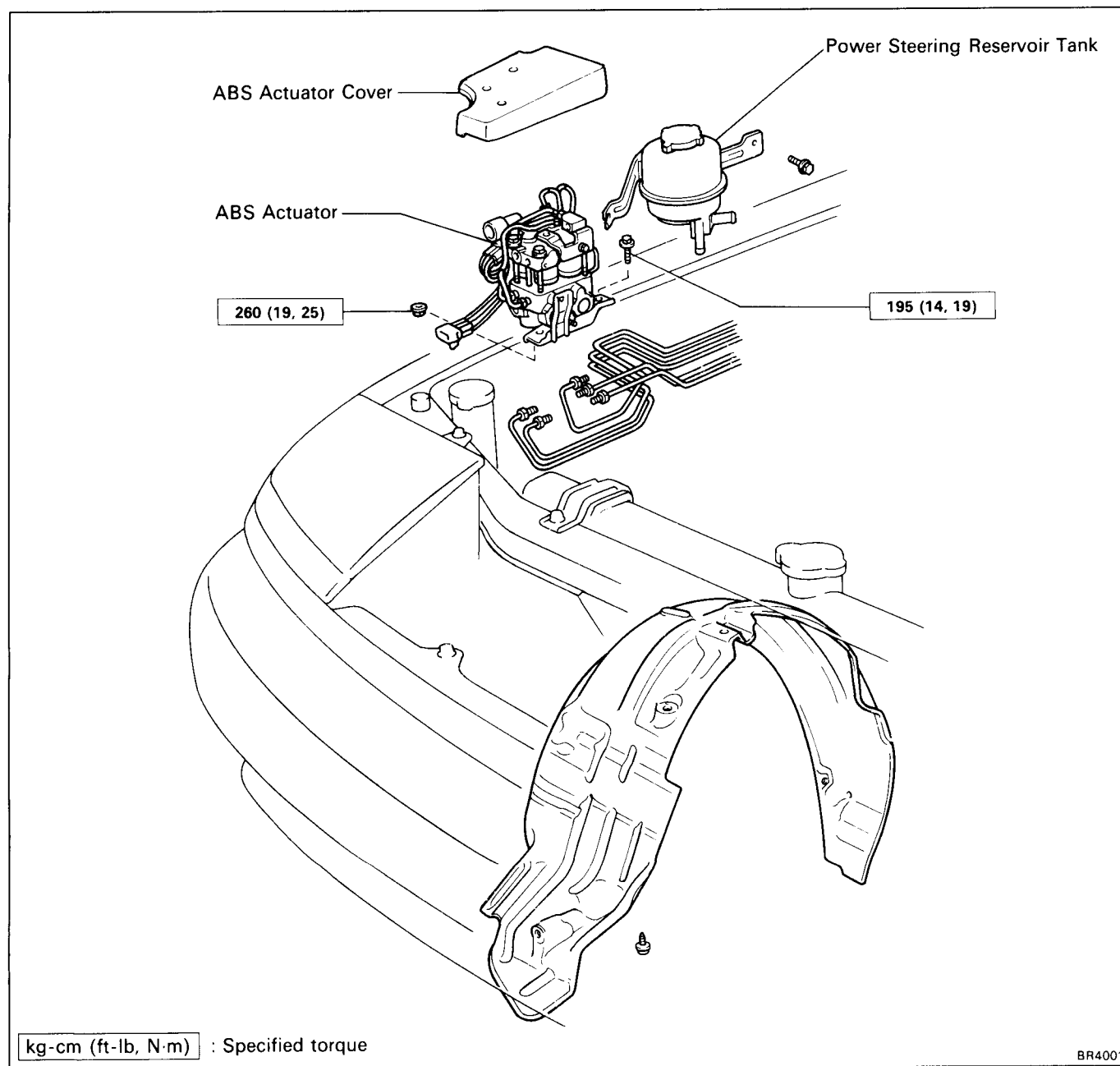
- Stop the engine.
- Remove the SST from terminals Ts and E1 of the check connector.



ABS Actuator

REMOVAL AND INSTALLATION OF ABS ACTUATOR

Remove and install the parts as shown.



1. DISCONNECT AND CONNECT BRAKE TUBE

Using SST, disconnect and connect the brake tubes from/to the ABS actuator.

SST 09751-36011

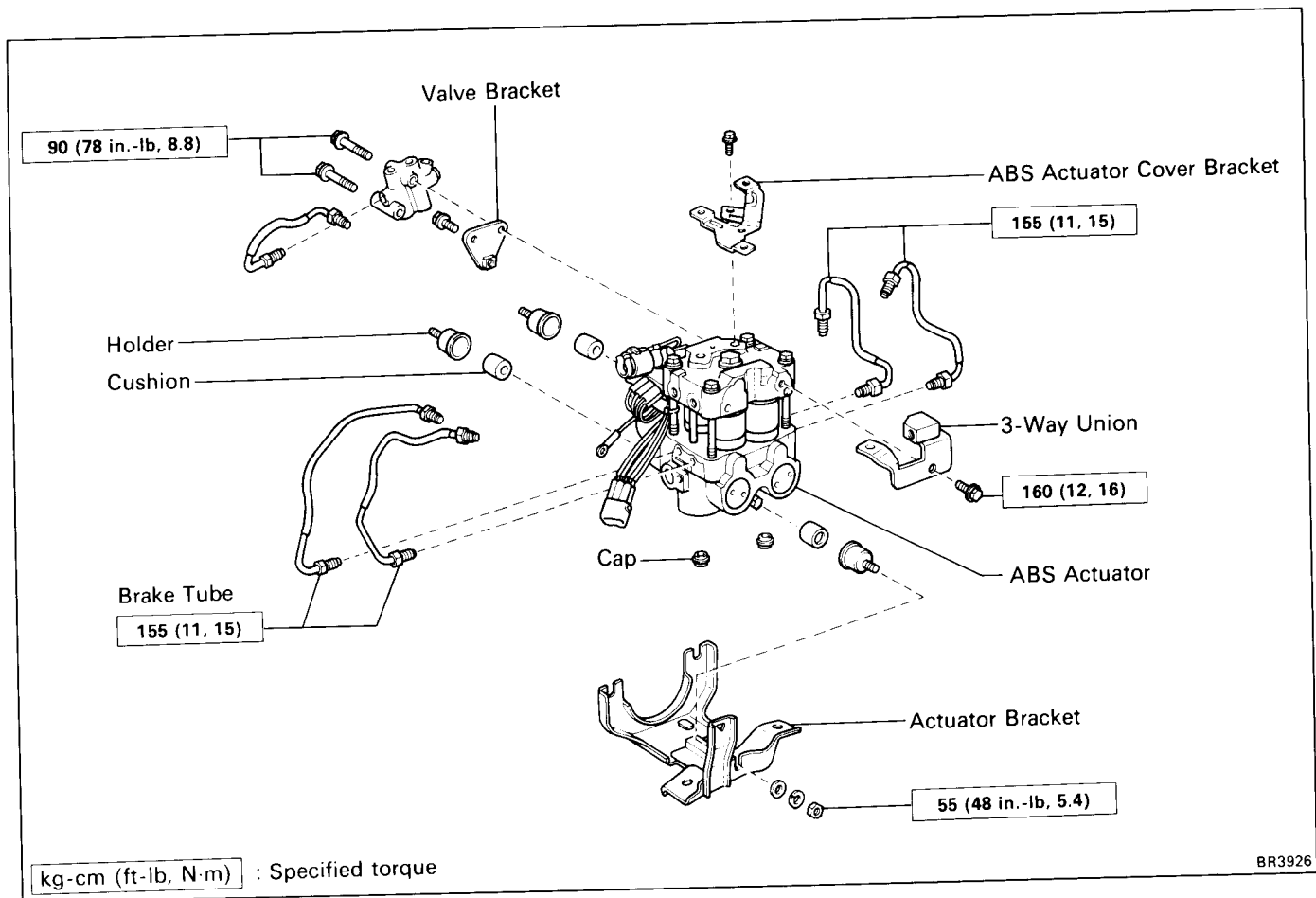
Torque: 155 kg-cm (11 ft-lb, 15 N·m)

2. FILL BRAKE RESERVOIR WITH BRAKE FLUID AND BLEED BRAKE SYSTEM

DISASSEMBLY AND ASSEMBLY OF ABS ACTUATOR

Remove and install the parts as shown.

COMPONENTS



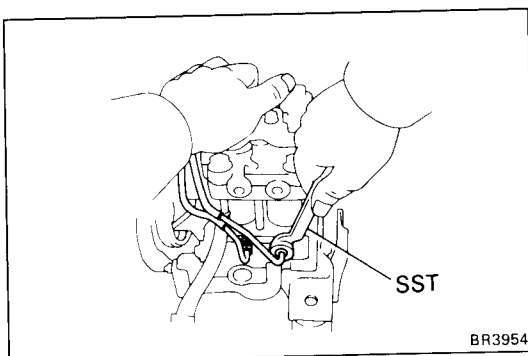
(MAIN POINT OF REMOVAL AND INSTALLATION)

1. DISCONNECT AND CONNECT BRAKE TUBE

Using SST, disconnect and connect the brake tubes from/to the ABS actuator.

SST 09751-36011

Torque: 155 kg-cm (11 ft-lb, 15 N·m)

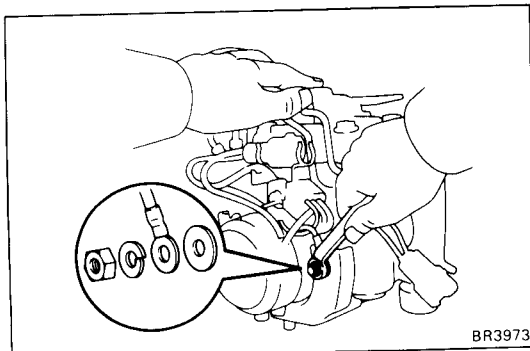


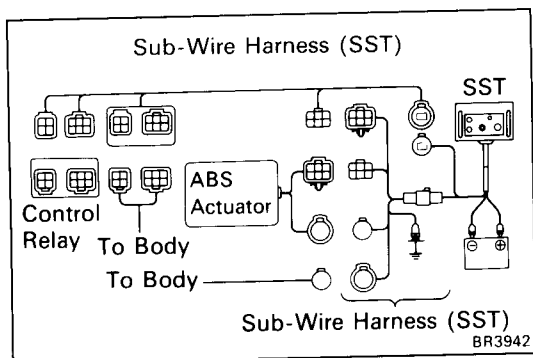
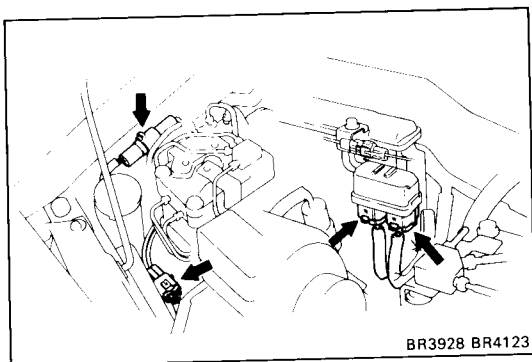
2. INSTALL ACTUATOR

- Install the actuator to the actuator bracket in place.
- Install the washers, wave washers and nuts.
- Tighten the three nuts.

Torque: 55 kg-cm (48 in.-lb, 5.4 N·m)

HINT: Before installing the front right side nut, install the motor ground terminal between the plate washer and spring washer.





INSPECTION OF ABS ACTUATOR OPERATION

1. INSPECT BATTERY VOLTAGE

Battery voltage: 10 – 14.5 V

2. REMOVE ACTUATOR COVER

3. DISCONNECT CONNECTORS

Disconnect the four connectors from the actuator and control relay.

4. CONNECT ACTUATOR CHECKER (SST) TO ACTUATOR

- (a) Connect the actuator checker (SST) to the actuator, control relay and body side wire harness through the sub-wire harness (SST) as shown.

SST 09990-00150 and 09990-00200

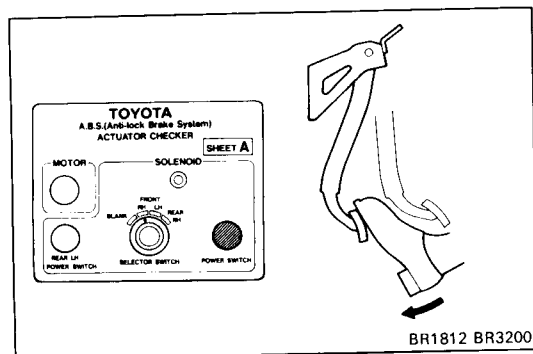
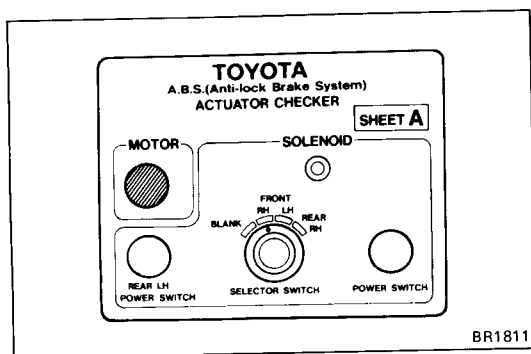
- (b) Connect the red cable of the checker to the battery positive (+) terminal and black cable to the negative (-) terminal. Connect the black cable of the sub-wire harness to the battery negative (-) terminal or body ground.

- (c) Place the "SHEET A" (SST) on the actuator checker.

SST 09990-00163

5. INSPECT BRAKE ACTUATOR OPERATION

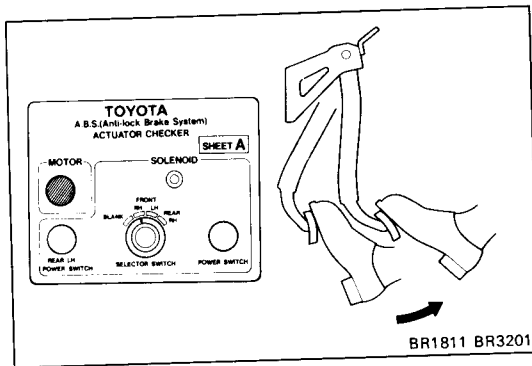
- (a) Start the engine, and run it at idle.
 (b) Turn the selector switch of the actuator checker to "FRONT RH" position.
 (c) Push and hold in the MOTOR switch for a few seconds.
 (d) Depress the brake pedal and hold it until the step (g) is completed.



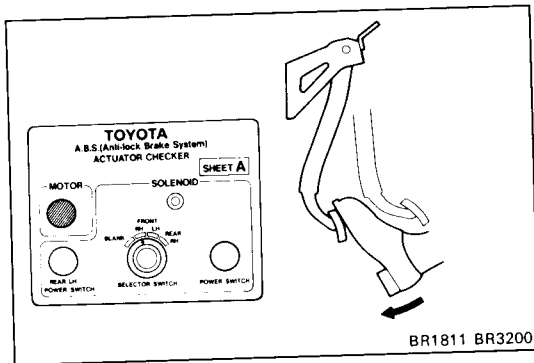
- (e) Push the POWER SWITCH, and check that the brake pedal does not go down.

NOTICE: Do not keep the POWER SWITCH pushing more than 10 seconds.

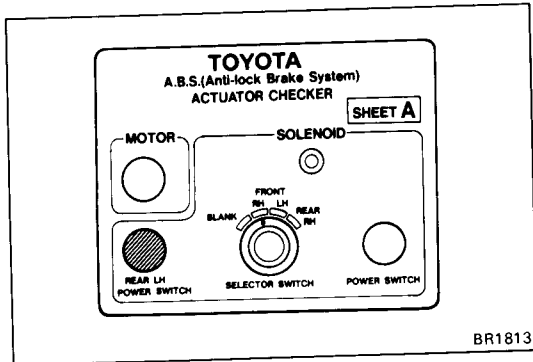
- (f) Release the switch, and check that the pedal go down.



- (g) Push and hold in the MOTOR switch for a few seconds, and check that the pedal returns.
- (h) Release the brake pedal.



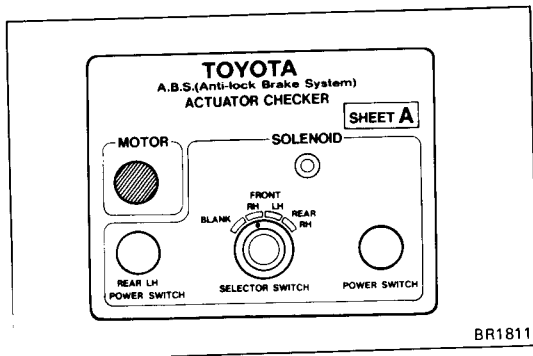
- (i) Push and hold in the MOTOR switch for a few seconds.
- (j) Depress the brake pedal and hold it for about 15 seconds. As you hold the pedal down, push the MOTOR switch for a few seconds. Check that the brake pedal does not pulsate.



6. INSPECT FOR OTHER WHEELS

- (a) Turn the selector switch to "FRONT LH" position.
- (b) Repeating (c) to (j) of the step 5, check the actuator operation similarly.
- (c) Similarly, inspect "REAR RH" and "REAR LH" position.

HINT: When inspecting "REAR LH" position, push the REAR LH switch instead of the POWER SWITCH, and you can inspect in any selector switch position.



7. PUSH MOTOR SWITCH

Push and hold in the MOTOR switch for a few seconds.

8. DISCONNECT ACTUATOR CHECKER (SST) FROM ACTUATOR

- (a) Remove the "SHEET A" (SST) and disconnect the actuator checker (SST) and sub-wire harness (SST) from the actuator, control relay and body side wire harness.

SST 09990-00150, 09990-00200 and 09990-00163

9. CONNECT ACTUATOR CONNECTORS

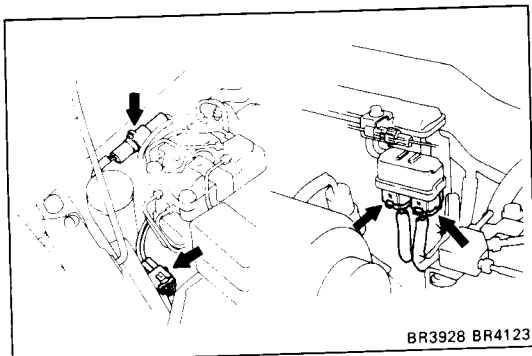
Connect the four connectors to actuator and control relay.

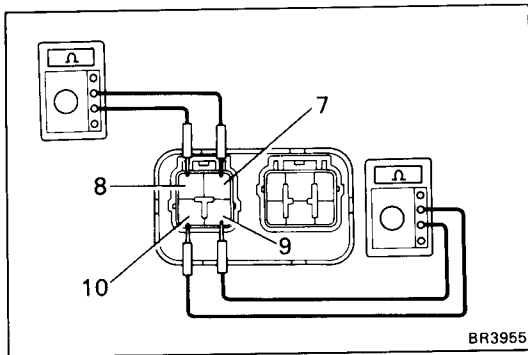
10. INSTALL CONNECTOR

Install the connector to the actuator bracket.

11. INSTALL ACTUATOR COVER

12. CLEAR DIAGNOSTIC CODES (See page BR-9)





Control Relay

INSPECTION OF CONTROL RELAY

1. INSPECT CONTINUITY OF MOTOR RELAY CIRCUIT

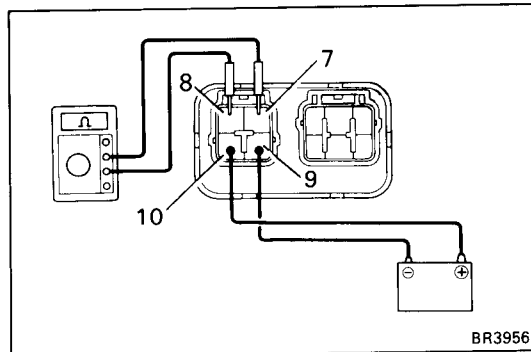
- Check that there is continuity between terminals 9 and 10.
- Check that there is no continuity between terminals 7 and 8.

If continuity is not as specified, replace the relay.

2. INSPECT OPERATION OF MOTOR RELAY CIRCUIT

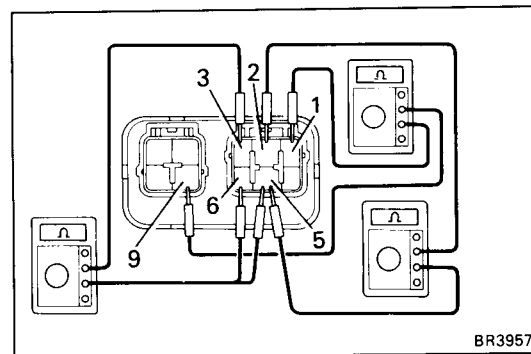
- Connect the positive (+) lead from the battery to terminal 10 and negative (-) lead to terminal 9.
- Check that there is continuity between terminals 7 and 8.

If operation is not as specified, replace the relay.



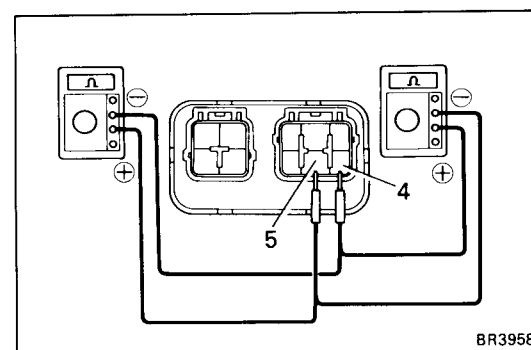
3. INSPECT CONTINUITY OF SOLENOID RELAY CIRCUIT

- Check that there is continuity between terminals 1 and 9.
- Check that there is no continuity between terminals 2 and 5.
- Check that there is continuity between terminals 3 and 5 or 6.



- Connect the positive lead from the ohmmeter to terminal 5 and connect negative lead to terminal 4. Check that there is continuity between terminals.
- Connect the two leads in reverse, and check that there is no continuity between terminals.

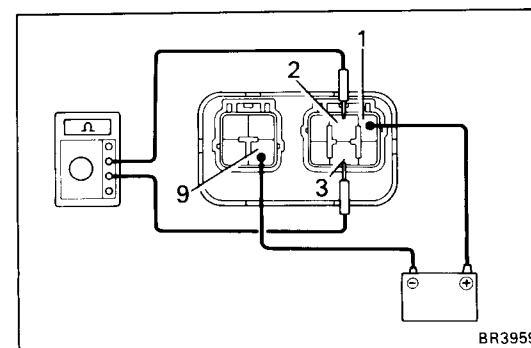
If continuity is not as specified, replace the relay.



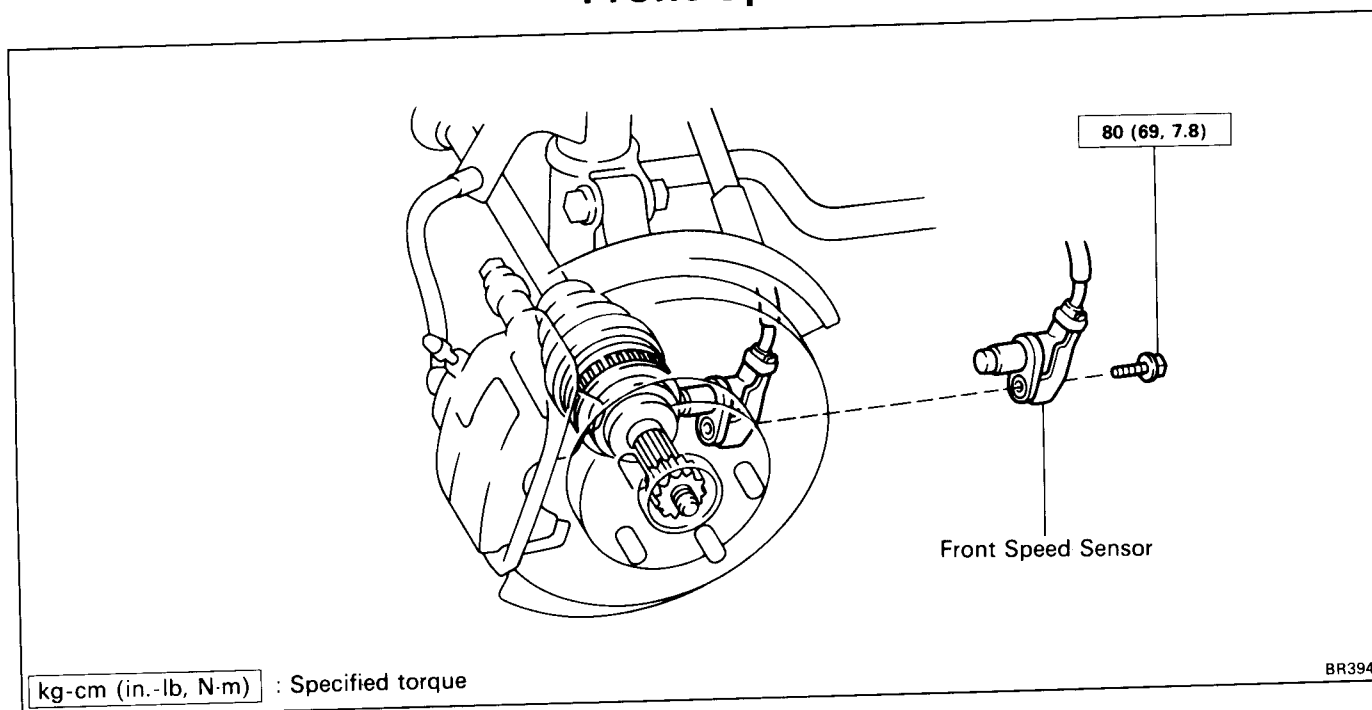
4. INSPECT OPERATION OF SOLENOID RELAY CIRCUIT

- Connect the positive (+) lead from the battery to terminal 1 and negative (-) lead to terminal 9.
- Check that there is continuity between terminals 2 and 5.

If operation is not as specified, replace the relay.



Front Speed Sensor



INSPECTION OF FRONT SPEED SENSOR

1. INSPECT SPEED SENSOR

- Remove the bolt from the pipe clamp of the wire harness.
- Disconnect the speed sensor connector.
- Measure the resistance between terminals.

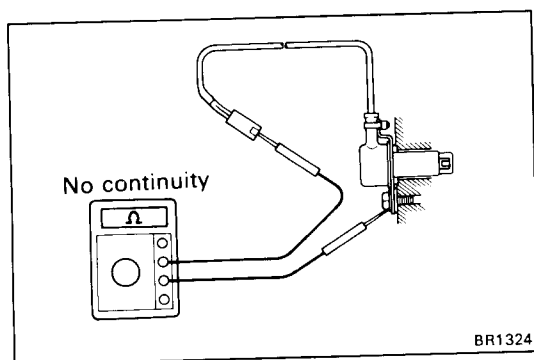
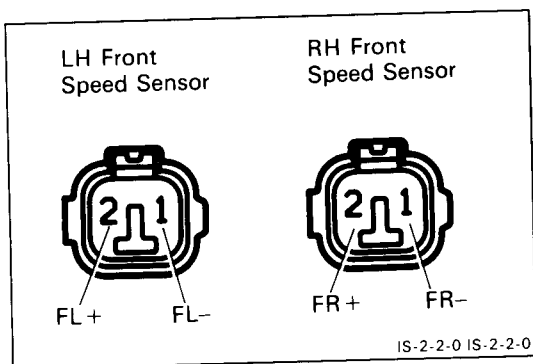
Resistance: 0.8 – 1.3 kΩ

If resistance value is not as specified, replace the sensor.

- Check that there is no continuity between each terminal and sensor body.

If there is continuity, replace the sensor.

- Connect the speed sensor connector.
- Install the bolt of the pipe clamp.



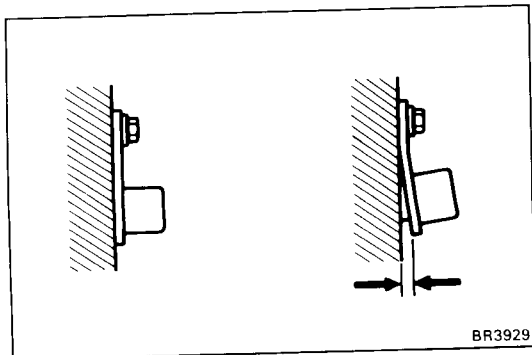
2. INSPECT SENSOR INSTALLATION

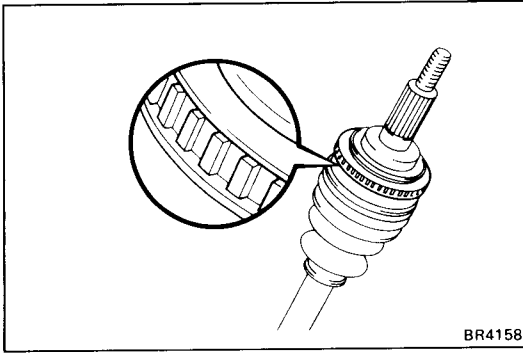
- Check that the sensor installation bolt is tightened properly. If not, tighten the bolt.

Torque: 80 kg-cm (69 in.-lb, 7.8 N·m)

- Check that there is no clearance between the sensor and rear axle carrier as shown.

If there is clearance, replace the sensor.

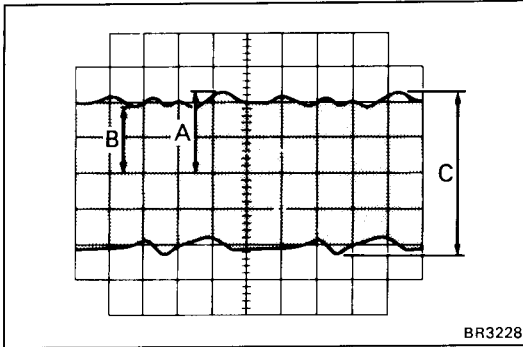




3. VISUALLY INSPECT SENSOR ROTOR SERRATIONS

- (a) Remove the drive shaft.
(See page SA-17 or 42)
- (b) Inspect the sensor rotor serrations for scratches, cracks, warping or missing teeth.
- (c) Install the drive shaft.
(See page SA-17 or 42)

NOTICE: To prevent damage to the serrations, do not strike the drive shaft.



INSPECTION OF FRONT SPEED SENSOR AND SENSOR ROTOR SERRATIONS (REFERENCE)

INSPECT FRONT SPEED SENSOR AND SENSOR ROTOR SERRATIONS BY USING AN OSCILLOSCOPE

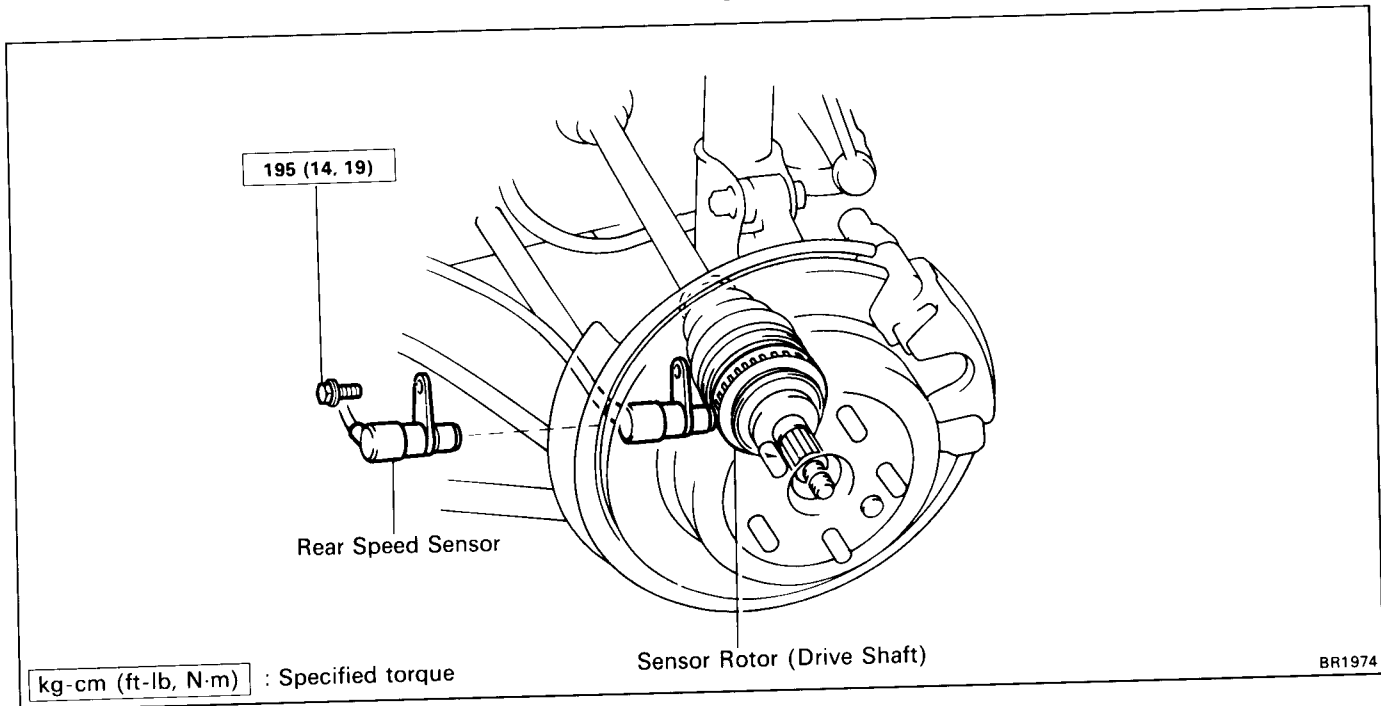
- (a) Connect an oscilloscope to the speed sensor connector.
- (b) Run the vehicle at 20 km/h (12.4 mph), and inspect speed sensor output wave.
- (c) Check that C is 0.5 V or more.

If not as specified, replace the speed sensor.

- (d) Check that B is 70 % or more of A.

If not as specified, replace the drive shaft.

Rear Speed Sensor



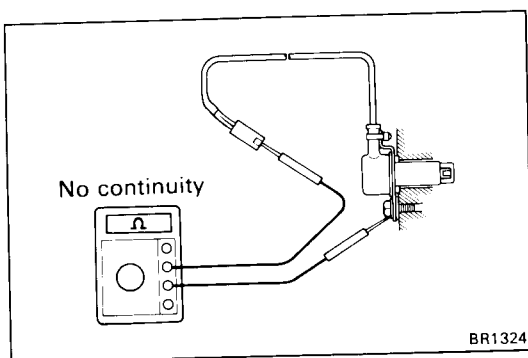
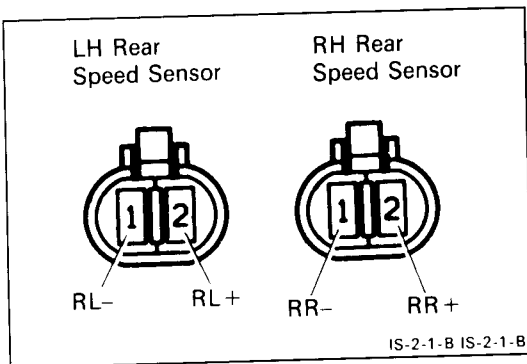
INSPECTION OF REAR SPEED SENSOR

1. INSPECT SPEED SENSOR

- Remove the seat cushion.
- Disconnect the speed sensor connector.
- Measure the resistance between terminals.

Resistance: 0.8 – 1.5 K Ω

If resistance value is not as specified, replace the sensor.



- Check that there is no continuity between each terminal and sensor body.

If there is continuity, replace the sensor.

- Connect the speed sensor connector.
- Install the seat cushion.

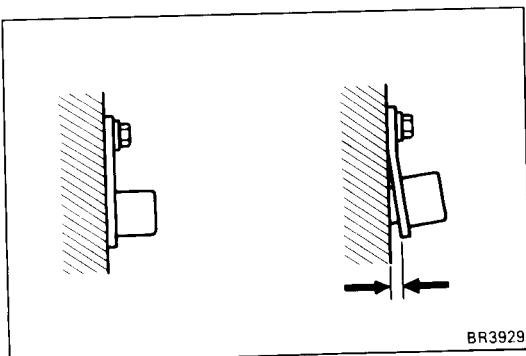
2. INSPECT SENSOR INSTALLATION

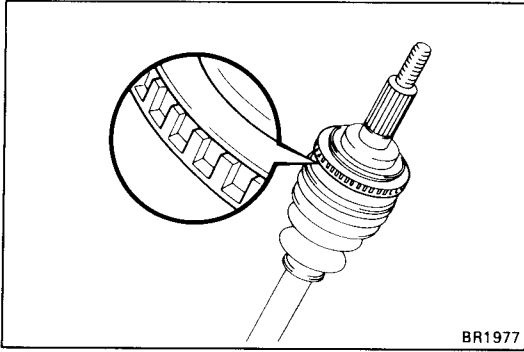
- Check that the sensor installation bolt is tightened properly. If not, tighten the bolt.

Torque: 195 kg-cm (14 ft.-lb, 19 N·m)

- Check that there is no clearance between the sensor and rear axle carrier as shown.

If there is clearance, replace the sensor.

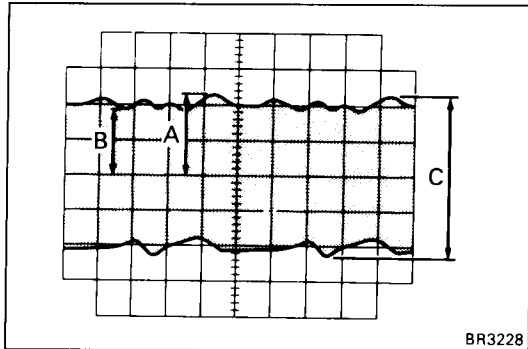




3. VISUALLY INSPECT SENSOR ROTOR SERRATIONS

- (a) Remove the drive shaft.
(See page SA-87)
- (b) Inspect the sensor rotor serrations for scratches, cracks, warping or missing teeth.
- (c) Install the drive shaft.
(See page SA-87)

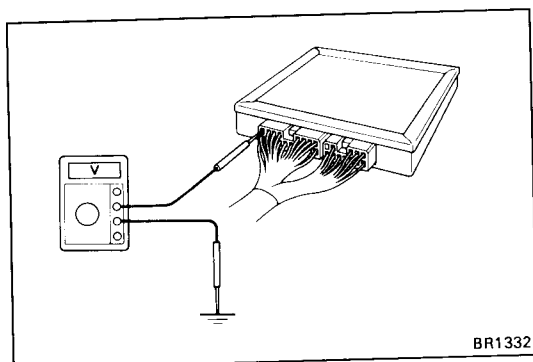
NOTICE: To prevent damage to the serrations, do not strike the drive shaft.



INSPECTION OF REAR SPEED SENSOR AND SENSOR ROTOR SERRATIONS (REFERENCE)

INSPECT REAR SPEED SENSOR AND SENSOR ROTOR SERRATIONS BY USING AN OSCILLOSCOPE

- (a) Connect an oscilloscope to the speed sensor connector.
- (b) Run the vehicle at 20 km/h (12.4 mph), and inspect speed sensor output wave.
- (c) Check that C is 0.5 V or more.
If not as specified, replace the speed sensor.
- (d) Check that B is 60 % or more of A.
If not as specified, replace the drive shaft.

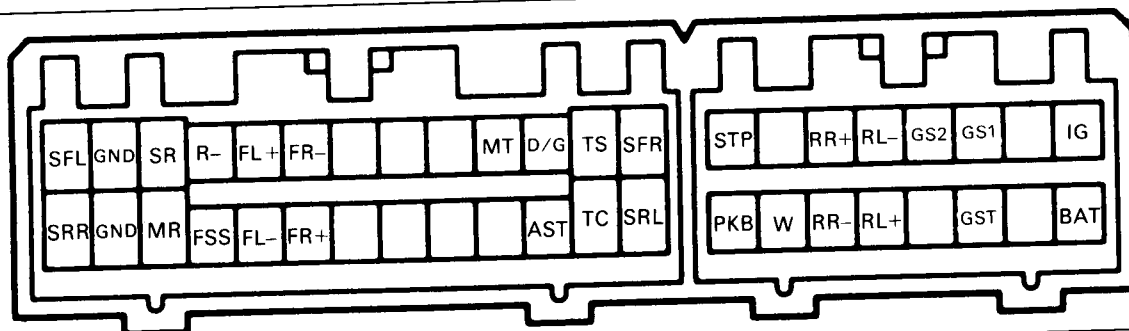


Anti-Lock Brake System Circuit

INSPECTION OF SYSTEM CIRCUIT

1. INSPECT SYSTEM CIRCUIT WITH CONNECTOR CONNECTED

- Remove the ABS ECU.
- Using a voltmeter with high impedance (10 K ohm/V minimum), measure the voltage at each terminal and body ground.



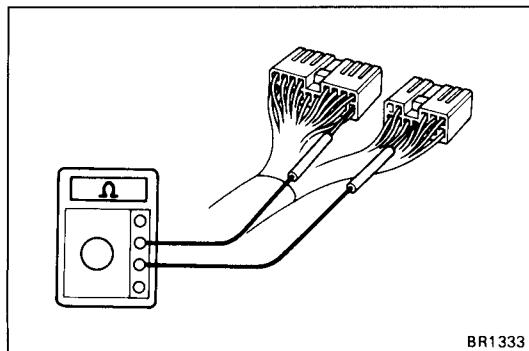
Tester Connection	Check Item	Condition	Specified Value	Trouble Part
IG	Voltage	IG switch on	Battery voltage	ECU-IG Fuse
GS1	Voltage	IG switch on	4 ~ 6 V	Deceleration Sensor
GS2	Voltage	IG switch on	4 ~ 6 V	
RL-	Continuity	IG switch off	Continuity	ABS ECU
STP	Voltage	IG switch off and brake pedal depressed	Battery voltage	Stop light switch Stop light
	Continuity	IG switch off and brake pedal returned	Continuity	
BAT	Voltage	IG switch off	Battery voltage	DOM E Fuse
RR-	Continuity	IG switch off	Continuity	ABS ECU
W	Voltage	IG switch on and "ABS" warning light goes on	About 0V	ABS ECU "ABS" warning light
		IG switch on and "ABS" warning light goes off	Battery voltage	
PKB	Voltage	IG switch on and PKB lever pulled	About 0V	Parking brake switch Level warning switch
		IG switch on and PKB lever returned	Battery voltage	
SFR	Voltage	IG switch on and "ABS" warning light goes on	About 0V	Actuator
		IG switch on and "ABS" warning light goes off	Battery voltage	
TS	Voltage	IG switch on and check connector Ts-E ₁ not connected	Battery voltage	ABS ECU
		IG switch on and check connector Ts-E ₁ connected	About 0V	
D/G	Voltage	IG switch on and check connector Ts-E ₁ not connected	About 0V	
FR-	Continuity	IG switch off	Continuity	ABS ECU
R-	Continuity	IG switch off	Continuity	
SR	Voltage	IG switch on and "ABS" warning light goes on	About 0V	
		IG switch on and "ABS" warning light goes off	Battery voltage	
GND	Continuity	IG switch off	Continuity	Wiring harness
SFL	Voltage	IG switch on and "ABS" warning light goes on	About 0V	Actuator
		IG switch on and "ABS" warning light goes off	Battery voltage	

Continued on page BR-32

Continued from page BR-31

Tester Connection	Check Item	Condition	Specified Value	Trouble Part
SRL	Voltage	IG switch on and "ABS" warning light goes on	About 0V	Actuator
		IG switch on and "ABS" warning light goes off	Battery voltage	
TC	Voltage	IG switch on and check connector Ts-E ₁ not connected	Battery voltage	ABS ECU
		IG switch on and check connector Ts-E ₁ connected	About 0V	
AST	Voltage	IG switch on and "ABS" warning light goes on	About 0V	Actuator
		IG switch on and "ABS" warning light goes off	Battery voltage	
FL-	Continuity	IG switch off	Continuity	ABS ECU
FSS	Continuity	IG switch off	Continuity	
SRR	Voltage	IG switch on and "ABS" warning light goes on	About 0V	Actuator
		IG switch on and "ABS" warning light goes off	Battery voltage	

If the circuit is not as specified, check and repair or replace the trouble part shown in the table above.



2. INSPECT SYSTEM CIRCUIT WITH CONNECTOR DISCONNECTED

- (a) Disconnect the connectors from the ECU, inspect at the wire harness side connector.

Tester Connection	Check Item	Specified Value	Trouble Part	Tester Connection	Check Item	Specified Value	Trouble Part
RR+ ↔ RR-	Resistance	0.8 ~ 1.5 kΩ	Rear RH speed sensor	SFL ↔ AST	Resistance	About 6 Ω	Actuator
RL+ ↔ RL-	Resistance	0.8 ~ 1.5 kΩ	Rear LH speed sensor	SRL ↔ AST	Resistance	About 6 Ω	Actuator
SFR ↔ AST	Resistance	About 6 Ω	Actuator	AST ↔ Body ground	Resistance	About 5 Ω	Actuator
MT ↔ Body ground	Continuity	Continuity	Actuator	FR+ ↔ FR-	Resistance	0.8 ~ 1.3 Ω	Front RH speed sensor
FL+ ↔ FL-	Resistance	0.8 ~ 1.3 kΩ	Front LH speed sensor	MR ↔ R-	Resistance	50 ~ 80 Ω	Control relay
SR ↔ R-	Resistance	60 ~ 100 Ω	Control relay	SRR ↔ AST	Resistance	About 6 Ω	Actuator

If the circuit is not as specified, check and repair or replace the trouble part shown in the table above.

- (b) Connect the connectors, and install the ECU in place.